ENGINEERING ENTREPRENEURSHIP AND IPR

MODULE 1

MODULE 1

- **Introduction to ideation, Innovation & Entrepreneurship:** What is Ideation? Understanding Innovation, Frameworks for Innovation, The Entrepreneurial for Innovation, The Entrepreneurial Mindset, Starting a Business, types formation statutory compliances
- **Introduction to Intellectual Property rights (IPR):** Types of IPR: Patents, trademarks, copyrights, trade secrets.

Strategies for protecting intellectual property based on the type of innovation

Entrepreneur vs Businessman vs Industrialist

Entrepreneur vs Businessman vs Industrialist

- Creates **new ideas**
- **High risk**
- Focus: **Innovation & growth**

N. R. Narayana Murthy

- Runs an **existing business**
- **Moderate risk**
- Focus: **Profit & stability**

Mukesh Ambani

- Manages **large industries**
- **High investment**
- Focus: **Mass production**

Ratan Tata

Entrepreneurship

- Entrepreneurship is about finding a solution to a problem or an opportunity.
- Business often revolves around identifying and addressing unmet needs or challenges in the market.
- Successful businesses provide solutions that solve problems for their customers, creating value and driving growth
- Example: Google Pay, Ola Cabs, Uber India, Swiggy and Zomato etc

Introduction to Ideation, Innovation and Enterpreneurship

[Ideas] → [Innovation Process] → [NewProduct/Service/Method]

What is Ideation?

- Ideation refers to the process of developing and conveying prescriptive ideas to others, typically in a business setting.
- It describes the sequence of thoughts, from the original concept to implementation.
- Ideation can be expressed in graphical, written, or verbal terms.
- Ideation is usually derived from brainstorming sessions, online forums, seminars, surveys, social media platforms, and team-building exercises.
- Anyone from an organization, from the CEO to an intern, can partake in the ideation process and contribute innovatively to a company.
- Most of the ideation process originates from trying to fix a problem; ideation is commonly reverse engineered.
- Styles of ideation include problem solutions, derivative ideas, and symbiotic ideas

Invention: Creating something entirely new that did not previously exist. Inventions are usually technological or mechanical creations.

> Example: The IC engine is an invention that transformed transportation. Prior to its invention, cars or other automobiles did not exist, and it changed the way people moved and travelled.

Discovery: Finding something that already exists but was unknown until it was found. Discovery often occurs in nature or science.

> Example: Electricity was discovered, not invented. Michael Faraday's experiments led to a deeper understanding of electromagnetic fields, which laid the groundwork for harnessing electricity.

Innovation: Taking an existing idea or product and improving it, making it more valuable or usable and being able to commercialise it. Innovation is often a refinement or application of an invention or idea.

> Example: The iPhone represents a significant innovation. While mobile phones existed, Apple redefined the smartphone market with a touchscreen interface, internet access, and apps.

Creativity: The mental ability to come up with original ideas or find unique solutions. Creativity often precedes invention and innovation and is the spark of novel thinking.

> Example: LEGO is a highly creative toy. The original bricks were created to encourage imaginative play, allowing children (and adults) to build endless configurations from their own minds.

Understanding Innovation

- Innovation is a key factor in entrepreneurship, as it helps entrepreneurs create new products and services that meet market needs.
- Innovation can also help entrepreneurs improve existing products, diversify their businesses, and stay ahead of competitors

- **Identify new opportunities**
- **Create new products and services**
- **Improve existing products**
- **Diversify businesses**
- **Stay ahead of competitors**
- **Foster creativity**

Barriers to innovation in entrepreneurship

1. **Lack of resources**
2. **Risk aversion**
3. **Lack of diversity**
4. **Bureaucracy**
5. **Resistance to change**
6. **Lack of employee recognition**

Types of Innovation

INNOVATION

Sustaining innovation

enhances an organization's processes and technologies to improve its product line for an existing customer base

Disruptive innovation

occurs when smaller companies challenge larger businesses

Architectural innovation

modification of existing technologies for use in entirely new markets

Eg mRNA technology platforms(think COVID vaccine),being used to develop treatments in multiple disease areas

Radical Innovation

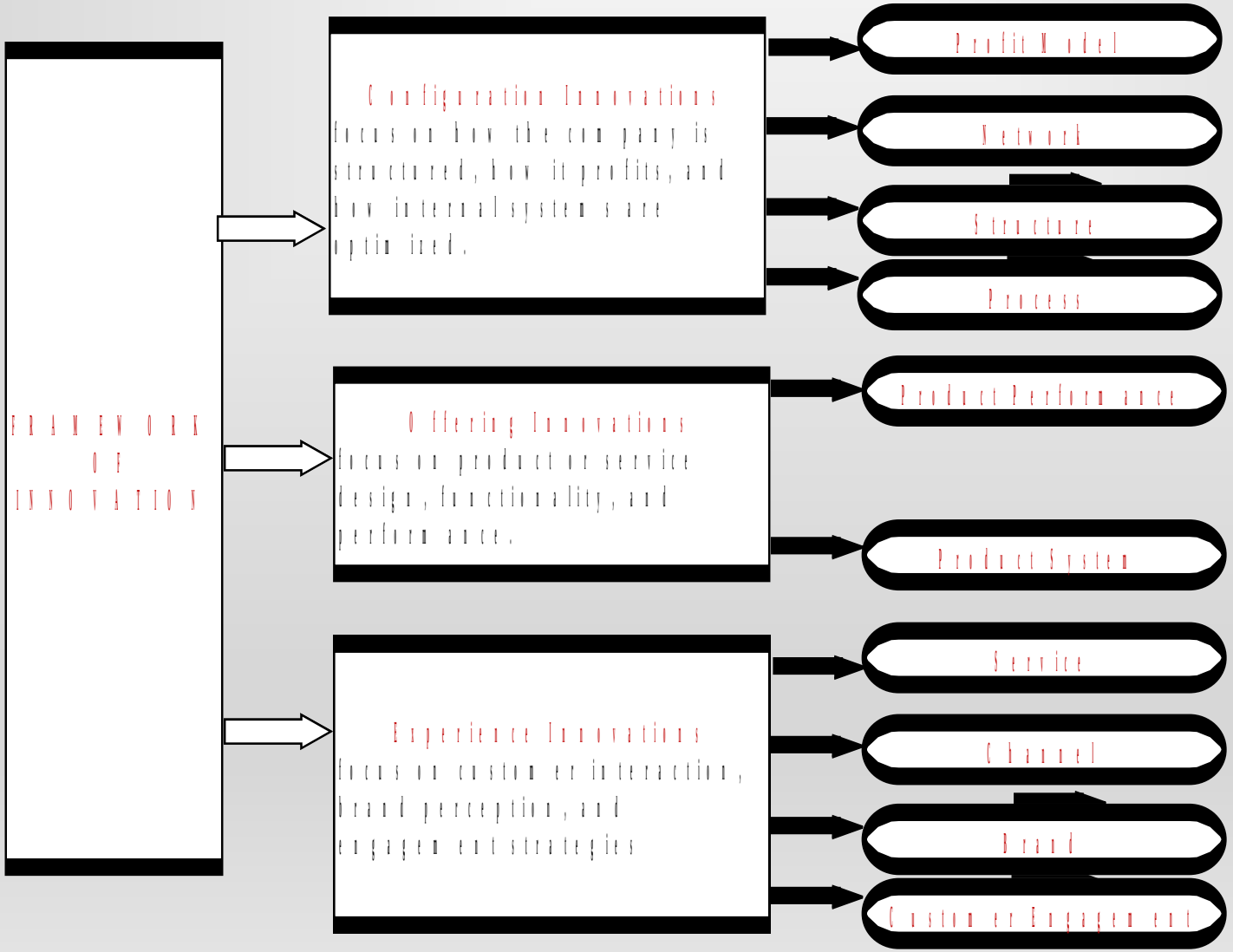
occurs when an organisation targets new markets with new technologies

Eg Introduction and use of x-rays, or the use of genome sequencing

Frameworks for Innovation

- An innovation framework, or innovation management framework, is a structured system designed to guide the entire process of innovation within a company.
- It outlines the steps and methodologies needed to develop, evaluate, and implement new ideas, ensuring they align with the organization's overall strategic objectives.
- This framework serves as a blueprint for how a company approaches innovation, from idea generation to the successful launch of new products or processes.

Types of Innovation Framework



1. Profit Model
Refers to innovative ways a company makes money.

Examples: Subscription models (Netflix), freemium models (Spotify), dynamic pricing (Uber).
2. Network
Leveraging partnerships and external relationships for competitive advantage.
Examples: Open innovation platforms (Android OS), supplier collaborations (Nike and Apple for smart footwear).

3. Structure
Optimizing organizational alignment to create value.
Examples: Decentralized workforce (GitLab), employee-driven innovation (Google's 20%-time policy).

4. Process
Improving core processes for efficiency, speed, or effectiveness.
Examples: Toyota's Lean Manufacturing, Zara's fast fashion supply chain.

5. Product Performance
Enhancing product capabilities, quality, or features.
Examples: Tesla's over-the-air software updates, Apple's retina display.

6. Product System
Creating complementary products/services that enhance value.
Examples: Apple's ecosystem (iPhone, Mac, iCloud), Microsoft Office Suite.

7. Service
Improving customer support, convenience, and after-sales experiences.
Examples: Zappos’ legendary customer service, Amazon Prime’s one-day delivery.

8. Channel
Innovative ways of reaching customers and delivering products/services.
Examples: Direct-to-consumer model (Warby Parker), online marketplaces (Airbnb, Amazon).

9. Brand
Establishing strong emotional connections and brand identity.
Examples: Nike’s "Just Do It" campaign, Tesla’s sustainability branding.

10. Customer Engagement
Enhancing interactions, personalization, and loyalty programs.
Examples: Starbucks Rewards, Netflix’s AI-driven recommendations.

Case Study: Amazon
Profit Model: Amazon Prime's subscription-based model.
Network: AWS partnerships with enterprises.
Process: Automated logistics and AI-driven recommendations.
Product System: Echo, Kindle, and Fire TV interconnected ecosystem.
Channel: E-commerce, physical stores (Amazon Go), voice commerce (Alexa).
Customer Engagement: Personalized recommendations and 1-click purchasing.

The Key Components of an Innovation Framework

- **1. Strategy and Vision**
- **2.Process and Methodology**
- **3. Open Framework**
- **4. Design Thinking Framework**

The Entrepreneurial Mindset

Some characteristics of an entrepreneurial mindset include

- Resilience
- Adaptability
- Risk-taking
- Continuous learning
- Solution-oriented
- Optimistic
- Authenticity

Starting a Business

- **The Sixteen Steps to Entrepreneurship**

Step 1: Identify Your Passions and Suitable Ideas

- **Core:** Discover business concepts that align with your skills, interests, and passions.
- **Action:** Brainstorm a variety of ideas, considering your knowledge, experience, and learning. Select ideas that you feel confident executing and that fit your budget.

Step 2: Conduct Preliminary Research on Ideas

- **Core:** Explore each shortlisted idea in detail.
- **Action:** Research each idea thoroughly, including market analysis, competitor analysis, and feasibility studies. Understand the infrastructure, machinery, raw materials, market potential, challenges, competitors, production process, and required capital for each idea.

Step 3: Shortlist Ideas
Core: Narrow down your list of ideas based on the research conducted.
Action: Evaluate each idea based on the insights gained in Step 2. Some ideas might become less appealing, while others may seem more promising.

Step 4: Evaluate Shortlisted Ideas
Core: Assess the shortlisted ideas based on three key factors: desirability, feasibility, and viability.
Action: Analyze each idea's potential appeal to customers (desirability), the availability of technical resources (feasibility), and its financial sustainability (viability). Seek expert advice if needed.

Step 5: Select the Best Idea & develop a prototype or MVP to test your product or service in the market and gather feedback from early adopters
Core: Choose the most promising idea from your shortlisted options and developing prototype and MVP
Action: Consider the insights gained in Step 4 and select the idea with the greatest potential for success and developing prototype and MVP

Step 6: Prepare a Business Plan

Core: Create a detailed document outlining your business strategy and operations.
Action: Include sections such as: Problem the business addresses
Solution the business offers
Target audience
Market size and opportunities
Marketing strategies
Competitors
Business model
Revenue model
Financial budget

Step 7: Explore Funding Options and Subsidies
Core: Identify available loans, subsidies, interest rates, repayment terms, and eligibility criteria for your business.
Action: Research government schemes and financial institutions offering funding for your type of business.

Step 8: Create a List of Experts and Institutions
Core: Build a network of experts and institutions who can support your business.
Action: Compile a list of contacts for professionals in various fields, including machinery suppliers, raw material providers, technical experts, financial institutions, industry experts, and financial advisors.

Step 9: Determine Capital Structure
Core: Plan the financial structure of your business, including equity and debt.
Action: Decide on the appropriate mix of equity (your own investment) and debt (loans) to finance your business. Consult with financial experts if needed.

Step 10: Obtain Necessary Permits and Registrations
Core: Identify and obtain all required legal documents and approvals.
Action: Prepare a list of necessary permits and registrations based on your business type and location. Ensure timely completion to avoid legal issues.

Step 11: Choose a Business Ownership Structure & & build a team
Core: Decide on the legal structure of your business (e.g., sole proprietorship, partnership, LLP, private limited company).
Action: Consider factors like tax implications, liability, and future growth potential. Consult with legal experts.

Step 12: Select a Name (Trademark) and Logo
Core: Choose a unique and memorable name and design a visually appealing logo for your business.
Action: Consider global branding and ensure your name and logo are distinctive and easily recognizable.

Step 13: Protect Intellectual Property
Core: Register your trademark, logo, and any other intellectual property.
Action: Protect your business's unique assets to prevent unauthorized use and maintain your competitive advantage.

Step 14: Choose a Location
Core: Select a suitable location for your business based on factors like accessibility, infrastructure, and target market.
Action: Consider the specific requirements of your business and evaluate different locations in terms of cost, convenience, and potential for growth.

Step 15: Obtain Legal Permissions
Core: Secure all necessary permits and approvals to operate your business.
Action: Ensure compliance with local regulations and industry-specific requirements.

Step 16: Launch Your Venture
Core: Implement your business plan and start operations.
Action: Follow your well-defined plan and monitor your business's performance. Be prepared to adapt and make adjustments as needed

Formation of organization

	Definition:	Advantages:	Disadvantages:
Sole Proprietorship Eg Freelancer,Local shops	A business owned and operated by a single individual	❖ Easy to set up ❖ Minimal paperwork ❖ full control over decision-making	❖ Unlimited liability ❖ limited access to capital ❖ difficulty in attracting talent
Partnership Eg Law firm	• A business owned by two or more individuals. • known as partners • partnership deed	❖ Shared responsibility ❖ pooled resources ❖ access to a wider network	❖ Potential for disagreements ❖ unlimited liability for partners ❖ complex legal structure

Limited Liability Partnership (LLP) Eg Razorpay	A partnership where partners have limited liability for the debts and obligations of the firm.	❖ Limited liability ❖ flexibility in profit sharing ❖ ease of formation	❖ Higher registration costs ❖ more complex legal structure
Private Limited Company Eg Flipkart	A company with a limited number of shareholders, typically not more than 200.	❖ Limited liability ❖ easier access to capital ❖ separate legal entity	❖ More complex formation process ❖ higher compliance requirements
Public Limited Company Eg TCS Infosys	A company with a large number of shareholders and shares listed on a stock exchange.	❖ Easier access to capital ❖ enhanced credibility, wider ownership base	❖ Complex formation process ❖ higher compliance requirements ❖ potential for hostile takeovers

Cooperative Society Eg Handloom weavers' cooperatives, dairy farmers' cooperatives	A business owned and operated by a group of individuals with a common interest.	❖ Democratic decision-making ❖ shared ownership ❖ potential for social and economic benefits	❖ Slower decision-making ❖ limited access to capital

The choice of business form depends on various factors

- Ease of formation
- Capital or financial requirements
- Nature of liability
- Control
- Stability and continuity
- Flexibility to conduct operations
- Secrecy
- Legal aspect

What is Statutory Compliance?

- Statutory compliance is the system of laws within which businesses have to function.
- In India, this means numerous laws and regulations made by many government bodies.
- Statutory compliance primarily aims at enabling firms to make their fair contribution to employees' welfare while staying within legal bounds defined by respective authorities.
- The legal aspects of registering a business and setting up the right team structure are essential steps in the entrepreneurial journey.
- Proper registration ensures compliance with regulations and reduces legal risks.

Key Statutory Compliance Requirements in India:

1. Registration of forms of business & Corporate Laws Compliance

- No specific registration is required for starting a sole proprietorship firm, certain basic registrations are required to be obtained by a sole proprietorship firm for doing business.
- The basic registrations required by a sole proprietorship are-

➢ The proprietor needs to obtain the Registration Certificate under the Shops and Establishment Act of the state in which the business is located.

➢ The sole proprietorship should also register for GST if the business turnover exceeds Rs.20 lakh.

➢ The sole proprietorship can also register as a Micro, Small and Medium Enterprise (MSME) under MSME Act, though it is not mandatory, it is beneficial to be registered under the same.

❑ The Indian Partnership Act, 1932 governs and regulates partnership firms in India.

❑ The Companies Act, 2013: Governs the incorporation, regulation, and dissolution of companies in India.

❑ The Competition Act, 2002: Ensures fair competition in the market, prohibiting practices like abuse of dominant position and anti-competitive agreements.

2. Tax Compliance

- Income Tax Act, 1961: This law requires businesses to pay taxes from their income and also submit annual returns.
- Goods and Services Tax (GST): It is an all-inclusive tax on the production, sale as well as consumption of goods and services within India which removes several indirect taxes.
- Professional Tax: Levied by state governments on employees receiving salaries and professionals.

3. Labor Law Compliance

The Labour laws in India are there to safeguard the rights of the workers and ensure them justice in workplaces.

- Payment of Wages Act, 1936: It guarantees prompt payment of remuneration without any illegal deductions.
- The Factories Act, 1948: It provides for safety, health, and welfare measures for the working population inside factories.
- The Minimum Wages Act of 1948 sets the minimum pay rates for workers who are skilled and unskilled alike.
- The Employees' Provident Fund and Miscellaneous Provisions Act of 1952 creates insurance plans, pension funds, and provident funds.
- The 1948 Employees' State Insurance Act provides provisions for medical benefits and health insurance for workers.

4. Environmental Compliance

- The Environment Protection Act of 1986 includes provisions aimed at both improving and safeguarding the environment.
- Air (Prevention and Control of Pollution) Act, 1981: Controls and reduces air pollution.
- Water (Prevention and Control of Pollution) Act, 1974: Regulates water contamination to maintain the quality of water.

Industry-Specific Compliance

Every industry has its own set of regulations:

- Pharmaceutical Industry: It should follow the Drugs and Cosmetics Act, of 1940.
- Food Industry: Food is regulated by the Food Safety and Standards Act, of 2006.
- IT Industry: The Information Technology Act, of 2000 governs it.

Introduction to Intellectual Property Rights (IPR)

What are Intellectual Property Rights (IPR)?

- Intellectual property is an umbrella term for a set of intangible assets or assets that aren't physical in nature.
- Holders of intellectual property rights have a monopoly on the usage of property or items for a specified time period.
- IPR is a general term covering patents, copyright, trademark, industrial designs, geographical indications, layout design of integrated circuits, undisclosed information (trade secrets) and new plant varieties

Patents – It is used for protecting new inventions, ideas, or processes.

- Patent holders need to pay periodic government renewal fees.
- An approved patent is for a limited time period.
- For startups, patents are invaluable in protecting novel technologies, processes, or products, giving them a competitive advantage

Eg Apple's iPhone design

Copyrights – It protects the ideas, examples would be written works, music, art, etc.

- For startups often rely on copyrights to protect software code, creative content, or marketing materials, ensuring that their intellectual assets are not unlawfully copied or reproduced.

Eg J K Rowling's Harry Potter books

Trademarks – It is something that protects the symbols, colors, phrases, sounds, design etc.

Eg Apple Logo

Trade Secrets – It may be strategies, systems, formulas, or other confidential information of an organization that provides them a competitive advantage in the market.

Eg Google's search algorithm

Importance of building a strong team

- In today's competitive business environment, the ability of a company to harness the power of effective teamwork can be a critical differentiator.
- Team building is more than just a series of activities; it is about creating a culture of collaboration, trust, and mutual respect.
- A strong and well-balanced team is crucial to the success of any enterprise, as it brings together diverse skills, perspectives, and expertise that drive innovation and problem-solving.
- Successful companies have demonstrated how the right team dynamics can lead to extraordinary achievements.

TeamIndus, India's private aerospace startup aiming for a moon mission, structured their team strategically by bringing together space scientists, engineers, and business strategists, all working collaboratively toward a highly complex and ambitious goal. Their team-oriented approach enabled them to secure funding, build partnerships, and push the boundaries of space technology.

Another example is the founding team of **Google** where Larry Page and Sergey Brin leveraged their technical expertise while later bringing in leaders like Eric Schmidt to handle business operations, scaling Google into a global tech giant.

Building a strong team is important because it can:

- **Improve communication:** Effective communication is the most important factor in successful teamwork. Team building can help employees feel more comfortable communicating with each other.
- **Increase productivity:** Teams can be more productive and efficient. They can also help meet tight deadlines and make heavy workloads more manageable.
- **Improve morale:** Team building can help employees feel more engaged and excited about their work.
- **Build trust:** Teamwork can help build trust among team members.
- **Foster creativity and learning:** Teams can learn from each other's different skills and strengths.
- **Improve problem solving:** Teams can bring together people with different ways of thinking and ideas.
- **Reinforce company culture and values:** Teamwork can help improve company culture.

1. Identifying roles

When identifying roles in a strong team, you can consider the skills and strengths of each team member and assign roles that align with those strengths. This can lead to increased productivity and job satisfaction.

Here are some tips for identifying roles in a strong team:

- Consider strengths

Conduct a skills assessment or discuss with each team member to identify their areas of expertise and interest.

- Define roles clearly

Make sure everyone is clear about their roles and responsibilities. Write down the expectations so that people can reference them.

- Reassess roles

Periodically reassess and update roles as projects progress or organizational frameworks change.

Some examples of team roles include:

- **Leader:** Makes sure the team has clear objectives, ensures members are engaged, and plans and prioritizes tasks
- **Challenger:** Questions effectiveness, drives for results, and can look at problems and see more than one solution
- **Shaper:** Thrives on pressure and drives the team forward
- **Team worker:** Perceptive and diplomatic, listens and encourages cooperation
- **Monitor evaluator:** Strategic and discerning, makes objective judgments
- **Implementer:** Practical and efficient, turns ideas into action

2. Skill sets

Some skills that are important for a strong team include:

Communication

The ability to convey information effectively through speech or writing.

Active listening

The ability to focus on what others are saying and provide appropriate feedback.

Conflict resolution

The ability to address and resolve disagreements and conflicts within the team.

Leadership

The ability to guide and motivate team members to reach common goals.

Problem solving

The ability to identify potential sources of conflict and address them proactively.

Organization

The ability to organize to support teamwork and team development.

Other skills that are important for a strong team include:

Respect, Accountability, Delegation, Open-mindedness, Collaboration, Goal setting, Decision making, Interpersonal skills, Time management, and Growth mindset

3. Team dynamics

Team dynamics are the behaviors and psychological processes that occur within a team and affect their performance. Each personality type in a group can unconsciously influence the direction of the team's dynamics. A strong team has a variety of skills and a good balance of team dynamics, including:

- **Clear communication**: Effective communication is the foundation of a successful team. It helps align goals, resolve conflicts, and avoid misunderstandings.
- **Trust**: Team members should be able to rely on each other and feel comfortable sharing ideas.
- **Role clarity**: Each team member should understand their role and responsibilities.
- **Diversity**: A diverse team brings a range of perspectives and solutions.
- **Conflict resolution**: Teams should be able to resolve conflicts constructively.
- **Active listening**: Team members should be fully present and engaged when listening to others.
- **Psychological safety**: Team members should feel comfortable taking risks and being vulnerable.
- **Meaningful work**: Team members should feel a sense of purpose in their work.
- **Dependability**: Team members should be reliable and complete quality work on time.
- **Structure and clarity**: The team should have clear goals, plans, and roles

Identifying Pain Points and problem statement

- A pain point is the emotional impact of a problem or obstacle, while a problem statement is a summary of that pain point.
- A problem statement should be concise and actionable, and it provides guidance for a project's user experience and user interaction.
- Identifying pain points can help you improve the customer experience, reach your target audience more effectively, and position your business as the solution.

Here are some ways to identify pain points and problem statements:

Analyze data

Look at data and metrics that measure the performance and outcomes of a process. This can include quantitative data like time, cost, quality, and customer satisfaction, or qualitative data like comments, reviews, or complaints.

Conduct research

Use methods like user research, contextual inquiry, user journey mapping, empathy mapping, the five whys, field studies, customer support analysis, and product review analysis.

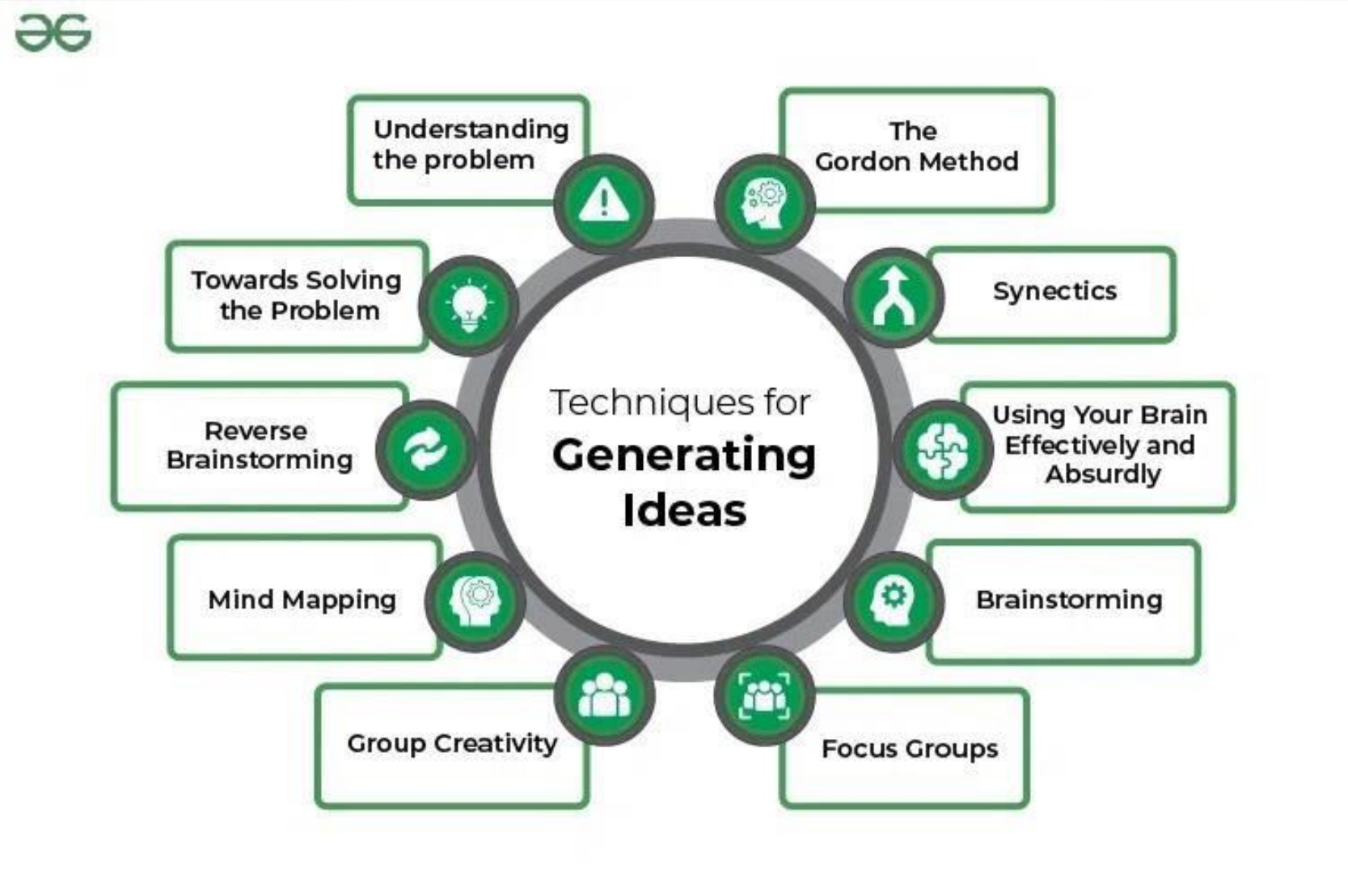
Create user personas

Use fictional representations of typical users to understand the needs and motivations of different user groups.

Listen to employees

Conduct surveys or have one-to-one conversations to learn more about your employees' experience.

Idea Generation Techniques



1. Understanding the Problem

Understanding the given problem is the most important step to solving the problem or generating ideas to solve the problem. It involves a systematic examination and comprehension of the problem the user is trying to solve. This is the first and the most crucial step in the case of creative problem-solving. The aim of this step is to lay a strong foundation that the ideas that will be generated later are well structured, informed, related to the main problem, and most likely to lead to the proper solutions. Some of the key characteristics of Understanding the problem are:

- **Problem Definition:** Before trying to solve the problem or even generate an idea, it is necessary to express the problem in the easiest way possible.
- **Gathering Information:** To understand the given problem, it is necessary to do some research and gather as much information as possible from other sources. This can include market research, data analysis, receiving feedback from customers, conducting surveys, etc.
- **Identification of Stakeholders:** It is also necessary to find out who will benefit from the solution of the problem, because they can provide some insights about the problem which will help in the idea generation process.

2. Towards Solving the Problem

The phase, Towards Solving the Problem, is the swift transition from understanding the problem to the phase of generating ideas. Once a deep understanding of the problem has been gained, the solver can start to think about it and generate ideas to solve the problem and think about the potential solutions.

- **Mind Mapping:** Mind Mapping is a visualization technique that is used to create a diagram like structure and put the ideas, tasks etc.
- **SCAMPER:** SCAMPER technique stands for Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, and Reverse is a technique which lets the user to view the problem from different angles and apply those aforementioned strategies to generate new ideas.
- **Lateral Thinking:** Lateral thinking is also another heavily used approach to generate ideas to solve problems. It seeks solution of the problem via indirect approaches.

3. Reverse Brainstorming

As the name itself suggests, Reverse Brainstorming is a well known problem solving technique which works in exactly opposite manner that of the traditional Brainstorming. Brainstorming sessions are usually done to generate ideas to solve a problem, whereas the reverse brainstorming is done to generate ideas which will cause a problem. Below are some of the key characteristics of Reverse Brainstorming:

- **Define the Problem:** It starts with defining the problem clearly and precisely, of whose solution we are trying to find.
- **Reverse the Goal:** Instead to directly finding the solution to the problem, encourage the participants to think how to create the problem or even make it worse. This approach will involve thinking about the actions, behavior and different factors which will lead to the problem.
- **Evaluate and Reverse:** After collecting a pool of ideas which causes the problem or make it harder, the participants evaluate those ideas, then starts the actual brainstorming method to find the solution of those problems in the traditional way.

4. Mind Mapping

Mind Mapping is a graphical and visualising approach used to represent the thoughts, ideas, concepts etc. and organise them in a hierarchical and interconnected manner. It is a very useful tool for brainstorming , problem solving, learning, planning and representing complex information. The main function of Mind Mapping is to recreate the process which happens inside human brain while trying to think about any solution of a particular problem. Below are the some of the most important components of a Mind Map:

- **Central Idea:** This occupies the center of the Mind Map, it represents the main concept or the main issue which is being solved.
- **Branches:** This are the thin lines that are used to join the key items and themes with the Central Idea in the middle.
- **Keywords or Phrases:** On each one of the branches (apart from the central idea), certain keywords or phrases are used to represent the main idea or issue of that sub-category or branch.

5. Group Creativity

Group Creativity is also known as collaborative creativity is a phase where a group of individual come together to share their ideas about solving a same problem. It is the ability of a team to work together to solve a certain problem given to them by thinking creatively and develop new concepts collectively to extend the capability of a single person's thinking power. Some of the most important points about Group Creativity is mentioned below:

- **Diversity of Perspectives:** Each group consists of members whose thinking capabilities are different than the others, they can come from different backgrounds, hold different experiences. This diverse atmosphere leads to more creative solutions.
- **Brainstorming:** Brainstorming are special sessions carried out to encourage the group members to encourage them and boost their creativity to generate more ideas. During this sessions the members are encouraged to share any kind of idea without any judgement or criticism.
- **Collaboration:** Collaboration and effective communication between the group members are a necessity for nurturing creative thinking.

6. Focus Groups

Focus Groups is a Qualitative research method which mostly involves a small group, but a diverse kind of individuals who freely shares their thought and opinions on a specific topic given to them. The discussions of a focus groups are generally mentored by a moderator or facilitator. Some key characteristics of Focus Groups are:

- **Size of the Group:** Generally the size of Focus Groups are small, it typically consists of only 6-10 individuals. It is kept like this because this is perfect number to get diverse ideas and keeping the encouragement and communication between them.
- **Structured Discussion:** As a certain moderator or Facilitator is in charge of a Focus Groups, he or she is in control of what should be discussed and what structure it would follow.
- **Interactive Discussion:** As the number of members are kept small, everyone gets the chance to participate in the discussion and interact with each other, this helps to discuss the ideas between them and rectify the mistakes if any.

7. Brainstorming

Brainstorming is a well known problem solving technique used to generate ideas within a group. Everyone in the group encourages each other to come up with new ideas whether they are absurd or practical without any criticism. The main goal of a brainstorm session is to come up with a solution of the problem or challenge given to them. Some of the characteristics of Brainstorming are:

- **Free Thinking:** All the members of the group can share their thoughts and ideas freely without facing any criticism. The main goal is to get as many idea as possible.
- **Divergent Thinking:** It is also a form of Divergent Thinking, it allows the members to think about multiple possible ways to solve the problem.
- **Done within a group:** Brainstorming sessions are usually conducted in a group. So the idea and the discussion remains secret to that group members only.

8. Using Your Brain Effectively and Absurdly

This encourages both rational thinking and the deliberate use of the unconventional concepts to stimulate creativity. It uses the both sides of a human mind, one that thinks rationally and tries to find the solution in a more rational manner, and the other which solely lives in the imaginary world and thinks absurdly. It is helpful to stimulate creativity in the mind's of the solver and generate ideas.

- **Effective Thinking:** This aspect of the using our brain effectively involves using the brain logically and efficiently. It solely depends upon the structured thinking and analysis of the problem thoroughly.
- **Absurd Thinking:** This might sound silly, but sometimes the absurd and imaginary thinking encourages the user to try some out of the box approaches to solve the problem.
- **Thinking Visually and Conceptually:** Using diagrams and Mind Maps is also helpful to effectively use our brain and generate new techniques to solve the problem.

9. Synectics

Synectics is used to establish connections between unrelated concepts/ideas generated during brainstorming sessions. It lets the users members to think beyond the traditional approaches and come up with something out of the box. Some of the key characteristics are:

- **Use of Analogy and Metaphors:** In Synectics, most of the times it relies on the use of different analogies and metaphors to establish connections between unrelated topics. By creating a connection between unrelated topics, new ideas can be generated easily.
- **Relaxation and Playfulness:** Creation of a Playful and Relax environment is necessary in case of Synectics. Relax environment is helpful to encourage people generate more ideas.
- **No Judgement or Criticism:** The participants are encouraged to not to judge others after listening to their ideas and not criticise them. This allows the other participants to think more freely without fearing of criticism.

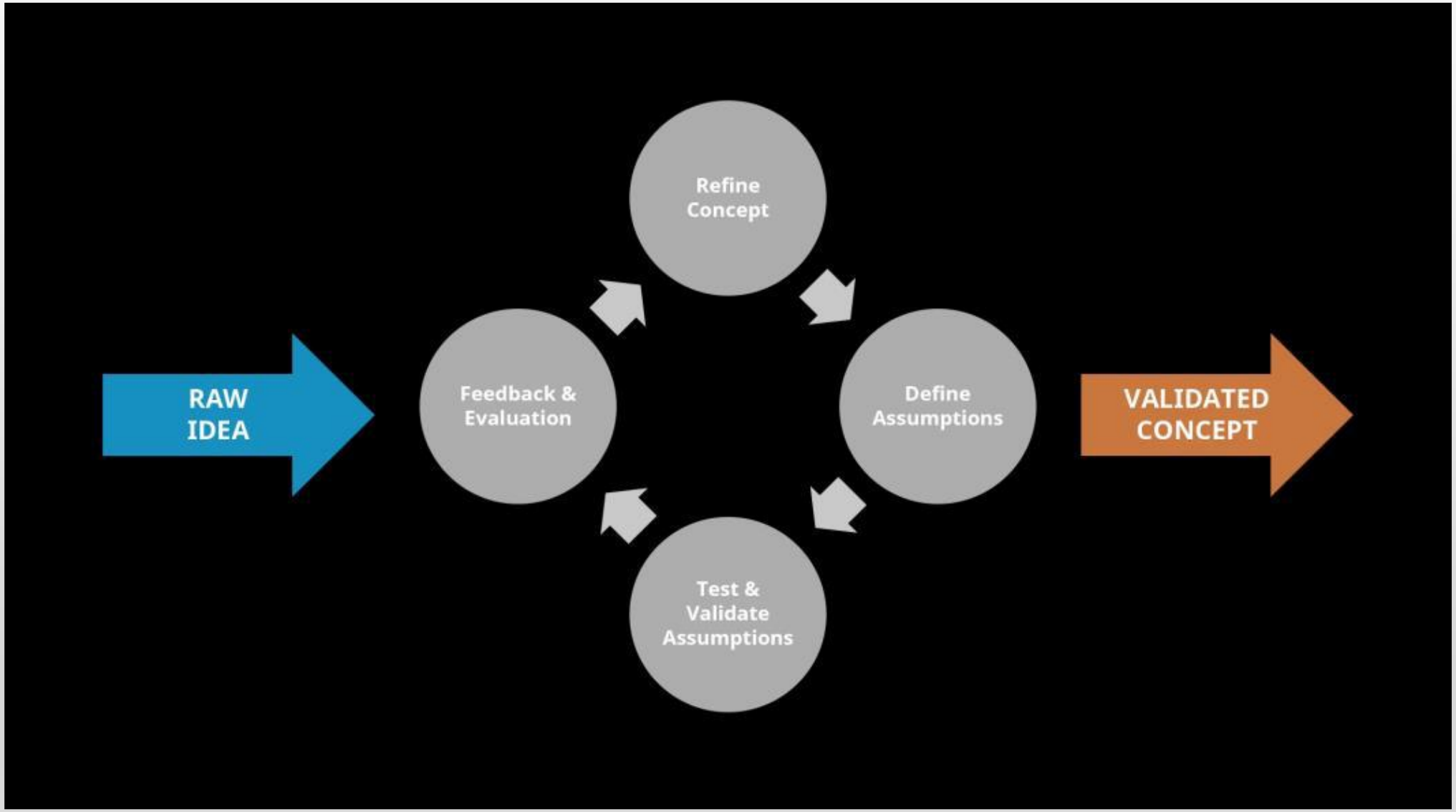
10. The Gordon Method

The Gordon Method is another well known problem solving approach developed by William JJ Gordon, this method is named after him. He was an engineer and a creativity expert. He is also known for its work in creating the Synectic method. The Gordon Method is closely related to the Synectic method, there are certain differences between them, like Synectic method is more structured than that of Gordon Method and it put emphasis on the use of Analogies and Metaphors to generate ideas. It follows a pre-defined process and uses specific "Operators" (Like images, diagrams etc as a hint to boost the idea generation), whereas the Gordon Method allows more flexibility in problem solving and relies upon the group collaboration, often with a Facilitator but no operators are used. Some of it's key characteristics are:

- **Clear Problem Statement:** The Gordon Method starts with a clear and concise Problem Statement which needs to be solved.
- **Emphasis on Organisational Goal:** Gordon Method plays an important role of putting Emphasis on the Organisation's goal and objectives which aligns with Creative Thinking.
- **Use of Triggers and Stimuli:** The Gordon Method uses various types of Triggers and Stimuli like specific structured questions, scenarios, or various challenges which can provoke creative ideas.

Developing and Refining Ideas

- Refining your ideas with feedback and data can help you validate assumptions, challenge biases, and discover new possibilities.
- Here are some strategies for developing and refining ideas:
- **Brainstorm:** Come up with multiple ideas and consider which is most relevant.
- **Get feedback:** Ask people with diverse perspectives to evaluate your idea objectively and critically.
- **Refine:** Use the feedback to make your idea more practical.
- **Test:** Test your idea in real or simulated situations to measure its performance and outcomes.
- **Collaborate:** Involve stakeholders to get a well-rounded perspective and address diverse needs.
- **Research:** Be prepared to modify your topic as you research. You might start with a broad topic and narrow it down, or vice versa.
- **Prototype:** Create a detailed business plan and construct a prototype of your idea. Early-stage prototypes can be as simple as a drawing or a computer render.


RAW IDEA
Refine Concept
Define Assumptions
Test & Validate Assumptions
Feedback & Evaluation
VALIDATED CONCEPT

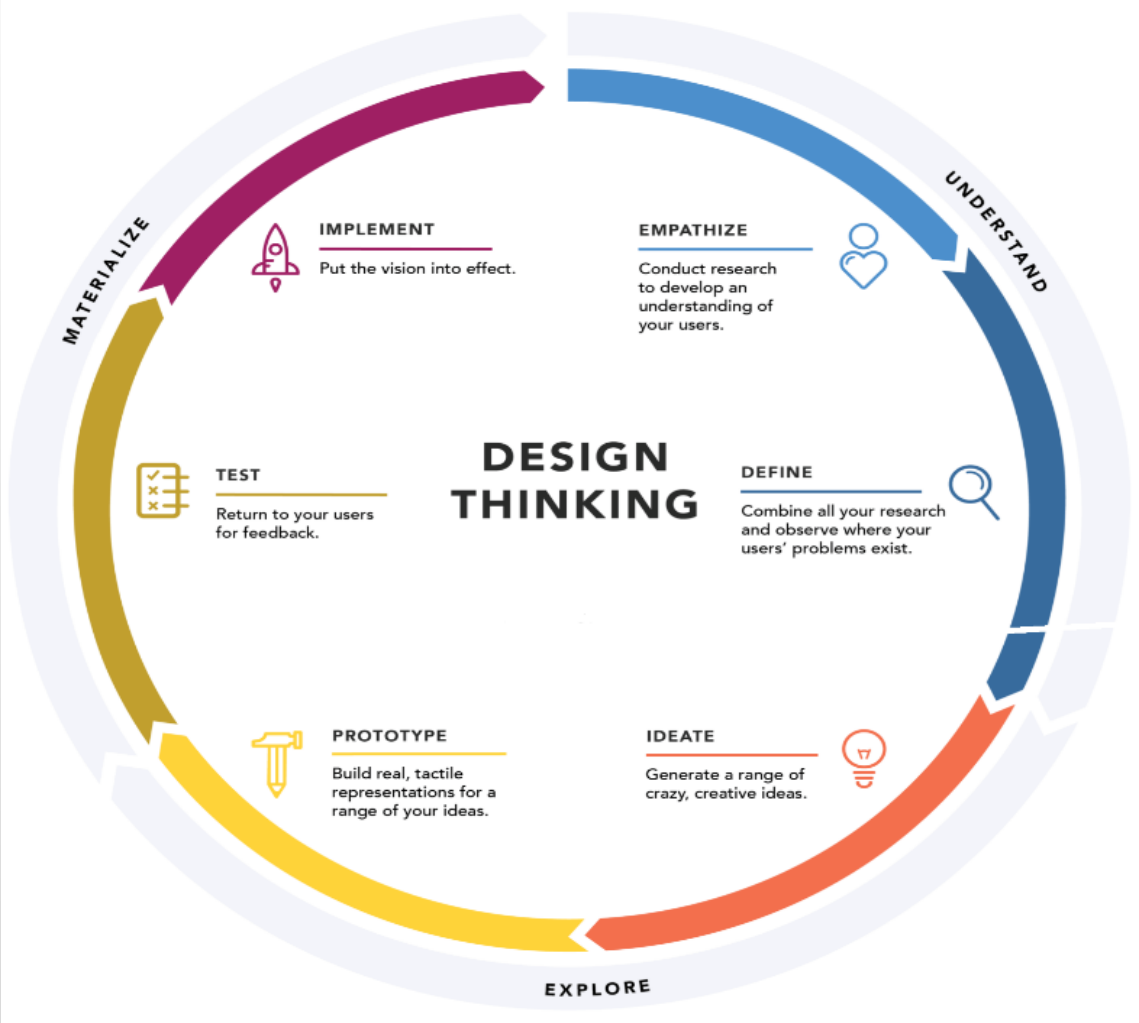
Develop strategies for bringing your innovation to life

- Innovation can help companies stand out, attract new customers, and deepen relationships with existing ones.
- Here are some strategies for bringing an innovation to life:

- **Define objectives**: Clearly define your innovation strategy and ensure it aligns with your company's goals.
- **Collaborate:** Encourage collaboration across departments to leverage different perspectives and expertise.
- **Make it practical:** Create a prototype, show examples, and get customer input.
- **Show customer benefit:** Demonstrate how your innovation will solve customer problems or concerns.
- **Build a business case:** Show how your innovation will add value, improve efficiency, or reduce costs.
- **Allocate resources:** Set aside time and resources for innovation, and budget for these initiatives.
- **Find partners:** Look for partners to co-develop your idea, especially if it's too complex for your team to handle alone.
- **Invite your leader:** Get your leader to co-own the initiative.
- **Consider timing:** Make sure the timing is right for your leader.

Design Thinking

- Design thinking is a mindset and approach to problem-solving and innovation anchored around human-centred design.
- The essence of design thinking is human-centric and user-specific.
- Design thinking is a non-linear, iterative process which consists of five stages
  - Empathise – Understand users' needs, challenges, and experiences through observation and interaction.
  - Define – Clearly articulate the problem based on insights gathered.
  - Ideate – Brainstorm potential solutions without constraints.
  - Prototype – Create simple versions of solutions to test and iterate.
  - Test – Evaluate the effectiveness of prototypes by gathering feedback and refining.



Design thinking is a human-centred approach to innovation that draws from the designer's toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success.

Design Thinking: Empathise

The Empathise phase is critical because it ensures that designers are addressing real user needs and creating solutions with the end-user in mind. The Empathise phase is essential because it lays the foundation for creating solutions that truly resonate with the users

Design Thinking: Analyze

The Analyze phase follows the Empathize phase and focuses on interpreting the insights gained from users to clearly define the problem that needs to be solved. The goal is to sift through data collected (interviews, surveys, observations) and determine the underlying challenges or patterns. By carefully analyzing this data, designers develop an actionable problem statement or point of view that drives solution ideation in the next phase.

Design Thinking: Ideation

Generate a wide range of creative ideas without judgment during the Ideation phase of Design Thinking and apply techniques and frameworks to explore multiple perspectives. Analyse potential solutions based on feasibility, user desirability, and business viability, and evaluate ideas to identify the most promising solutions for further development

Frameworks for Ideation
Brainstorming
SCAMPER Technique
Mind Mapping
Idea Matrix
Feasibility, Viability, Desirability (FVD) Analysis

Stakeholders

- A stakeholder is a person, group, organisation with vested interests in decision making and activities of a business or project

Internal stakeholders

- Employees
  - Individuals working with the organisation
- Owners
  - The individuals or entities that own the organisation
- Board of Directors
  - The governing body of organisation
- Project managers
  - Individuals responsible for overseeing the project
- Investors
  - Individuals or entities providing financial support to the project

External Stakeholders

- Suppliers
  - Organisation that provides goods or services to the project
- Customers
  - Individuals or organisations that purchase the project output
- creditors
  - Entities that have lent money to the organisation
- Clients
  - Individuals or organisations that have contracted with the organisation for the project
- Intermediaries
  - Entities that facilitate transactions between the organisations and other parties
- Competitors
  - Organisations that offer similar products or services
- Society
  - The boarder community that may be affected by the project's outcomes
- Government
  - Regulatory bodies or governments agencies with jurisdiction over the project